Project Main Problem: Dynamic Parking Assignment in Urban Networks

A driver is traveling through a city from a starting point to a final destination. Along the way, there are several parking lots near the destination where the driver could park. The challenge is to choose the best parking lot for the driver while they are still driving.

However, conditions in the city are always changing. Traffic can get better or worse, parking spots can fill up or become free, and parking costs can go up or down. This means that a parking lot that looked good at the beginning of the trip might become a bad choice by the time the driver gets closer.

Because of these changes, the parking decision cannot be made just once at the start. The system must be able to update its recommendation while the driver is moving, possibly suggesting a different parking lot as new information comes in. Furthermore, the driver has a personal limit for how far they are willing to walk. Any chosen parking lot must be within this maximum walking distance from the destination, or it will not be considered a valid option.

Current parking systems often pick lots based only on distance or price at one moment in time. They do not remember past mistakes or learn from experience, so they might keep sending drivers to lots that are often full, cause delays, or are beyond the driver's acceptable walking range.

The goal is to create a smarter system that learns over time. This system should consider driving time, walking distance from the parking lot, and parking cost altogether. It must also respect the driver's walking distance limit. The system should adjust its decisions as the environment changes, aiming to minimize the driver's total time and cost while ensuring the final walk is acceptable.

A good way to build such a system is by using reinforcement learning, where the system learns by trying different decisions and seeing the results, so it can make better and better choices in the future.

As an example, Figure 1 shows a system model of this problem. A vehicle is traveling from a source toward a destination. Two parking lots, P1 and P2, are located near the destination within the maximum walking distance. As the vehicle moves along the road, the system must dynamically evaluate and choose between P1 and P2 based on changing traffic, availability, cost, and the walking constraint.

The urban road network is modeled as a directed graph $\boldsymbol{G} = (\boldsymbol{N}, \boldsymbol{E})$. The node set $\boldsymbol{N}$ includes road intersections, the driver's origin $\boldsymbol{S}$, destination $\boldsymbol{D}$, and a set of parking lots $\boldsymbol{P} = \{\boldsymbol{p_1}, \boldsymbol{p_2}, \dots, \boldsymbol{p_m}\}$. Each parking lot $\boldsymbol{p_k}$ is associated with a parking cost $\boldsymbol{c_k}$, with the full set of costs denoted as $\boldsymbol{C} = \{\boldsymbol{c_1}, \boldsymbol{c_2}, \dots, \boldsymbol{c_m}\}$. The edge set $\boldsymbol{E}$ represents road segments, where each directed edge $\langle \boldsymbol{i}, \boldsymbol{j} \rangle \in \boldsymbol{E}$ connects two nodes. Every edge $\langle \boldsymbol{i}, \boldsymbol{j} \rangle$ is assigned a weight $\boldsymbol{T_{ij}}$ representing travel

time. To reflect dynamic traffic conditions, $\boldsymbol{T_{ij}}$ varies over time according to the simulation, even though the underlying network structure $\boldsymbol{G}$ remains fixed.

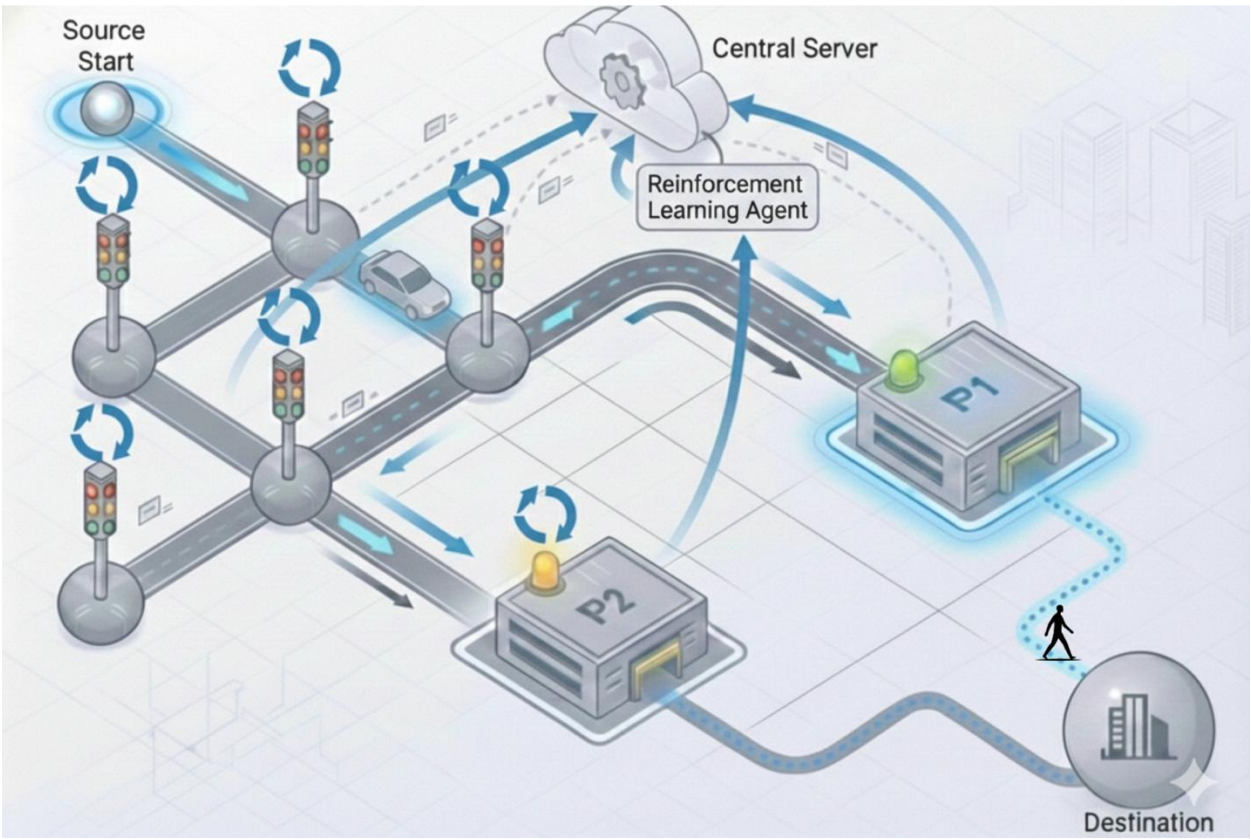

Figure 1. System Model of the Parking Assignment Problem